



On 3-component domination numbers in graphs

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ABSTRACT

Let s be a positive integer and let $G = (V(G), E(G))$ be a graph. A vertex set D is an s -component dominating set of G if every vertex outside D has a neighbor in D and every component of the subgraph induced by D in G contains at least s vertices. The minimum cardinality of an s -component dominating set of G is the s -component domination number $\gamma_s(G)$ of G . Determining the exact values or bounds of domination parameters on graphs is an important, basic, and challenging problem in the graph domination field. The tree T and the generalized Petersen graph $P(n, k)$ with $k \geq 1$ are the significant graph classes in graph theory. In this paper, we first give an upper bound of the 3-component domination number of a tree T . Then, we study the s -component domination numbers on $P(n, k)$ and get the exact values of 3-component domination numbers on $P(n, 1)$ and $P(n, 2)$.

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1. Introduction

Throughout this paper, only finite undirected and simple graphs are considered. A graph $G = (V, E)$ consists of a set V of vertices and a subset E of unordered pairs of vertices known as edges. The neighborhood of a vertex v , denoted $N_G(v)$, is defined as $\{u : uv \in E\}$. Additionally, $N_G[v]$ is defined as $N_G(v) \cup \{v\}$. The subgraph of G induced by a set S is represented as $G[S]$. The length of the shortest path between two given vertices u and v is called the *distance* between u and v , denoted by $d_G(u, v)$. The *diameter* of a graph G is the greatest distance between two vertices of G , denoted as $\text{diam}(G)$. We define the distance between two induced subgraphs G_1 and G_2 of G as $d(G_1, G_2) = \min_{v \in V(G_1), u \in V(G_2)} \{d_G(v, u)\}$. When there is no confusion in the context, we call a component with order s in some induced subgraph of G as s -component. For the definition of basic concepts not provided here, we recommend referring to a graph theory textbook such as [14].

A vertex set $S \subseteq V$ is considered to *dominate* V if every vertex in $V \setminus S$ is adjacent to at least one vertex in S . On the other hand, a *total dominating set* S requires that every vertex in V is adjacent to at least one vertex in S . The minimum cardinality of subsets S that dominate V is referred to as the *domination number* of G , denoted as $\gamma(G)$. Similarly, the total domination number is defined and denoted as $\gamma^t(G)$. The study of graph domination, initiated by Ore [20], has attracted much attention due to its practical applications [4,5,12,25]. To gain a comprehensive knowledge of this topic, we recommend some surveys [13,16] and excellent books [13–15,17] on dominating set and total dominating set in graphs. These references provide in-depth discussions and insights into the subject.

Separate results and proofs have been established for the domination number [9,18,19,21] and the total domination number [3,6,7]. To unify these results and generalize statements obtained for $\gamma(G)$ and $\gamma^t(G)$, Alvarado et al. [1,2] introduced a significant domination parameter—the s -component domination number $\gamma_s(G)$, which is stated as follows.

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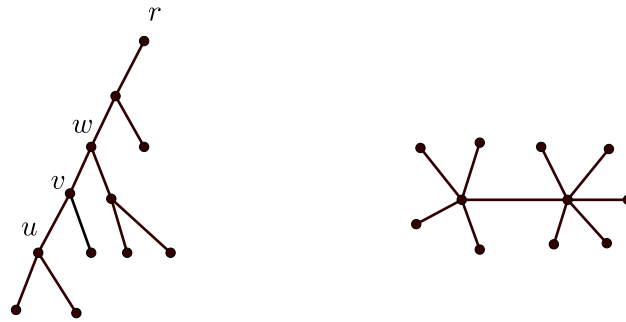


Fig. 1. A rooted tree and a double star.

Given a positive integer s , we call a vertex subset D is an s -component dominating set of G if it is a dominating set, and the order of every component in $G[D]$ is at least s . The minimum cardinality required for an s -component dominating set of G is denoted as the s -component domination number $\gamma_s(G)$. It is worth noting that $\gamma_1(G)$ coincides with the domination number of G , while $\gamma_2(G)$ coincides with the total domination number of G . In the following section, we denote an s -component dominating set with cardinality $\gamma_s(G)$ as a $\gamma_s(G)$ -set. If a graph G has an s -component dominating set, we say that G can be s -component dominated.

The generalized Petersen graph $P(n, k)$, defined by Watkins [23], is an important class of graphs. In $P(n, k)$, the vertex set is the union of $U = \{u_1, u_2, \dots, u_n\}$ and $V = \{v_1, v_2, \dots, v_n\}$, and the edge set consists of $\{u_i u_{i+1}, u_i v_i, v_i v_{i+k}\}$, where $1 \leq i \leq n$ and the subscripts are taken modulo n . For simplicity, the components consisting of vertices from U are defined as U -components, and V -components are defined similarly. Determining the exact values of the domination parameters for generalized Petersen graphs is an interesting and challenging problem that has been extensively studied in the literature [8,10,22,24,26].

In the following section, we first present an upper bound of $\gamma_3(T)$ for any tree T . Afterward, we will demonstrate the exact values of $\gamma_3(P(n, 1))$, $\gamma_3(P(n, 2))$, and $\gamma_4(P(n, 2))$.

2. Results on trees

Given a tree T , a vertex of degree exactly 1 in T is a *leaf*, and a *supporting vertex* is a vertex with a leaf as its neighbor. A vertex v in T is called a *connected vertex* if it is adjacent to some supporting vertex, not a leaf. Now let us start with three straightforward observations.

Observation 1. If v is a supporting vertex of T , then v is contained in any $\gamma_3(T)$ -set of T .

Observation 2. If T is a tree with diameter of at least 5, then there exists a $\gamma_3(T)$ -set in T which contains no leaves of T .

Observation 3. If v is a connected vertex of T , then v is contained in any $\gamma_3(T)$ -set of T .

Since the 3-component domination number on paths can be derived from the definition, which was given in [11] stated in the following lemma.

Lemma 4 ([11]). Let P_n be a path. Then

$$\gamma_3(P_n) = \gamma_3(C_n) = \begin{cases} \frac{3n}{5}, & n \equiv 0 \pmod{5} \\ \frac{3n+2}{5}, & n \equiv 1 \pmod{5} \\ \frac{3n+4}{5}, & n \equiv 2 \pmod{5} \\ \frac{3n+6}{5}, & n \equiv 3 \pmod{5} \\ \frac{3n+3}{5}, & n \equiv 4 \pmod{5} \end{cases}$$

Theorem 5. If T is a tree of order $n (\geq 5)$ with $t (\geq 3)$ supporting vertices, then $\gamma_3(T) \leq \frac{3(n+t-1)}{5}$.

Proof. It is trivial when $\text{diam}(T) \in \{2, 3\}$. For the former case, the graph is a star, and the latter one is a double star (see Fig. 1). In both cases, $\gamma_3(T) = 3 < \frac{3(n+t-1)}{5}$. If $\text{diam}(T) = 4$, by Observations 1 and 3, we know that there is a 3-component dominating set D in T , which consists of all the supporting vertices and the connected vertex. So $|D| \leq t + 1$. Support

that T has ℓ leaves, then $n = t + \ell + 1 \geq 2t + 1$ and further $\frac{3(n+t-1)}{5} \geq \frac{3(2t+1+t-1)}{5} \geq t + 1$. Without loss of generality, we can assume that $\text{diam}(T) \geq 5$, and then $n \geq 7$.

We proceed by induction on n . The above analysis establishes the base cases. Assume that every tree T' of order $5 \leq n' < n$ with $t' (\geq 3)$ supporting vertices satisfies $\gamma_3(T') \leq \frac{3(n'+t'-1)}{5}$. Noting that if $t' = 2$, then T' is a path and we know that $\gamma_3(T') \leq \frac{3(n'+t')}{5}$. Let r be an end-vertex of a path with length $\text{diam}(T)$, and root T at r . Let u be a supporting vertex at the maximum distance from r , and let v be its parent in the rooted tree. We consider the following two cases.

Case 1. $\deg_T(v) \geq 3$

Now v is a supporting vertex or v has a child, besides u , as a supporting vertex. Let $T' = T - T_u$, where T_u is a set which contains u and its leaf neighbors in T . Then $n' = n - |N(u)| \leq n - 2$ and $t' = t - 1$. If T' is a path, then $\gamma_3(T') \leq \frac{3(n'+t')}{5}$. Otherwise, if $n' \leq 4$, then it is easy to check that $\gamma_3(T') = 3 \leq \frac{3(n'+t')}{5}$; if $n' \geq 5$, by induction we know that $\gamma_3(T') \leq \frac{3(n'+t'-1)}{5}$. Based on [Observations 1](#) and [2](#), there is a $\gamma_3(T')$ -set D' containing v in T . Adding u to D' is a 3-component dominating set for T . Thus, $\gamma_3(T) \leq \gamma_3(T') + 1 \leq \frac{3(n'+t')}{5} + 1 \leq \frac{3(n-2+t-1)}{5} + 1 < \frac{3(n+t-1)}{5}$. The result holds.

Case 2. $\deg_T(v) = 2$

Let w be the parent of v in T . Now we prove that w is contained in every $\gamma_3(T')$ -set in T . If w is a supporting vertex or a connected vertex, then this is true from [Observations 1](#) and [3](#). Otherwise, suppose w has a child as a connected vertex whose degree is at least 3, then it reduces to Case 1. Therefore, we can assume that all the children of w are connected vertices whose degree is 2. Since the order of every component of $\gamma_3(G)$ -set is at least 3, this implies the result holds.

Assuming that $\deg_T(w) \geq 3$. Let $T' = T - T_v$, where $T_v = v \cup T_u$. Then $n' = n - |N[u]| \leq n - 3$ and $t' = t - 1$. If $n' \leq 4$, since $\text{diam}(T) \geq 5$, the order of T' is exactly 4. Hence a 3-dominating set of order 4 for T is easily obtained. Now we can assume that $n' \geq 5$, and $t' = t - 1$. If T' is a path, then $\gamma_3(T') \leq \frac{3(n'+t')}{5}$ and $n' \leq n - 4$. Adding $\{u, v\}$ to the 3-dominating set of T' forms a 3-dominating set in T , thus $\gamma_3(T) \leq \gamma_3(T') + 2 \leq \frac{3(n-4+t-1)}{5} + 2 < \frac{3(n+t-1)}{5}$. Otherwise, based on the induction hypothesis $\gamma_3(T') \leq \frac{3(n'+t'-1)}{5}$. Adding $\{u, v\}$ to the 3-dominating set of T' forms a 3-dominating set in T . This implies that $\gamma_3(T) \leq \gamma_3(T') + 2 \leq \frac{3(n-3+t-1-1)}{5} + 2 < \frac{3(n+t-1)}{5}$.

Now we consider that $\deg_T(w) = 2$. Let $T' = T - T_w$, where $T_w = w \cup T_v$. If $n' \leq 4$, then a 3-dominating set of order 5 for T can be easily found since T is not a path. Now we assume that $n' = n - |N(u)| - 1 \geq 5$. Suppose T' is not a path. Then it is easy to check $t' \leq t$. More precisely, either $t' = t - 1$ or $t' = t$. For the former case, we get that $n' = n - |N(u)| - 1 \leq n - 4$. Adding $\{u, v, w\}$ to a 3-dominating set of T' forms a 3-dominating set of T , thus $\gamma_3(T) \leq \gamma_3(T') + 3 \leq \frac{3(n-4+t-1-1)}{5} + 3 \leq \frac{3(n+t-1)}{5}$. For the latter case, $t' = t - 1$ which means the deletion of w results in a new leaf. We name x as the new leaf in T' , which is a parent of w in T and $\deg_T(x) = 2$. Now let $T'' = T - T_x$, where $T_x = x \cup T_w$. Then $n'' = n - |N(v)| - 2 \leq n - 5$, $t'' \leq t$. Without loss of generality, we assume that $n' \geq 5$. Then $\gamma_3(T) \leq \gamma_3(T'') + 3 \leq \frac{3(n+s-1)}{5}$. Now T' is a path. Since T is not a tree, $n' \leq n - 5$ holds. Let $T'' = T - T_x$ where x is a neighbor of w not v and $T_x = x \cup T_w$. Thus $n'' \leq n - 6$. By [Lemma 4](#), adding $\{u, v, w\}$ to a 3-dominating set of T' forms a 3-dominating set of T , thus $\gamma_3(T) \leq \gamma_3(T') + 3 \leq \frac{3(n-6+t)}{5} + 3 \leq \frac{3(n+t-1)}{5}$.

From the above cases, the theorem holds. $\square$

Note that the bound in the above theorem is sharp. Let us consider a graph T is constructed from a star $T_{1,t}$ with $t \geq 3$ and make subdivisions on each edge in $T_{1,t}$ three times. It is easy to check that all the subdivision vertices form a $\gamma_3(T)$ -set, thus $\gamma_3(T) = \frac{3(3t+t+1+t-1)}{5} = 3t$.

Now we consider the lower bound for 3-component domination numbers of trees. Let T be a tree with ℓ leaves and t supporting vertices. We conjecture that $\gamma_3(T) \geq \frac{3(n-\ell+2)}{5}$. For support this result, we can see that if $\text{diam}(T) = 2$, then T is a star and $\gamma_3(T) = 3 > \frac{3(n-\ell+2)}{5}$; if $\text{diam}(T) = 3$, then T is a double star and $\gamma_3(T) = 3 > \frac{3(n-\ell+2)}{5}$; if $\text{diam}(T) = 4$, according to [observation 1](#) and [observation 3](#), we learn that $\gamma_3(T) = t + 1$. Because $n = t + \ell + 1$ and $t \geq \ell$, so it is obvious that $t + 1 \geq \frac{3(n-\ell+2)}{5}$. In conclusion, the lower bound $\frac{3(n-\ell+2)}{5}$ holds for the tree with diameter not exceeding 4.

Furthermore, we construct a class of trees whose 3-component domination number is precisely $\frac{3(n-\ell+2)}{5}$. To characterize the extremal trees that attain this bound, we define the tree family $\mathcal{T}$ as the set of all trees T that can be obtained from a sequence $T_1, T_2, \dots, T_k$ ($k \geq 1$) of trees. The initial tree T_1 is a path P_5 with supporting vertices x, y and a connected vertex z . The final tree in the sequence is denoted by $T = T_k$. For $k \geq 2$, T_{i+1} can be obtained recursively from T_i by one of the following operations. Let $A(T_1) = \{x, y, z\}$ and H be a path P_5 with supporting vertices u, v and a connected vertex w .

- Operation $\mathcal{O}_1$: Attach a vertex by adding an edge to any vertex of $A(T_i)$. Let $A(T_{i+1}) = A(T_i)$
- Operation $\mathcal{O}_2$: Attach a copy of H by adding an edge from a leaf of H to any leaf in T_i . Let $A(T_{i+1}) = A(T_i) \cup \{u, v, w\}$.
- Operation $\mathcal{O}_3$: Attach a copy of H by adding a new vertex w and edges uw and wz , where z is a leaf of T_i . Let $A(T_{i+1}) = A(T_i) \cup \{u, v, w\}$.

Theorem 6. If $T \in \mathcal{T}$ has ℓ leaves, then $A(T)$ is a $\gamma_3(T)$ -set of order $\frac{3(n-\ell+2)}{5}$.

Proof. According to the definition of $\mathcal{T}$, we can assume $T = T_k$. It is easy to verify that $A(T)$ is a 3-component dominating set of order $\frac{3(n-\ell+2)}{5}$. We prove by induction on the number of operations k , where it is straightforward to show that the

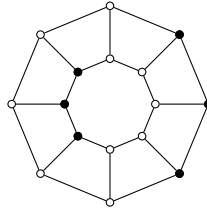


Fig. 2. Black vertices are chosen to be D and an 3-component dominating set for $P(8, 1)$.

conclusion holds for $k = 1$. Assuming the conclusion holds after $k - 1$ operations. Let D be a $\gamma_3(T)$ -set, and $T' = T_{k-1}$. Suppose T' contains n' vertices and ℓ' leaves.

If T is obtained from T' by operation $\mathcal{O}_1$, then obviously $A(T')$ is also a $\gamma_3(T)$ -set of T . Since $n = n' + 1$ and $\ell = \ell' + 1$, according to the induction hypothesis, $|A(T)| = \gamma_3(T)$.

Suppose T is obtained from T' by operation $\mathcal{O}_2$ or $\mathcal{O}_3$. Then $n = n' + 5$ and $\ell = \ell'$, or $n = n' + 6$ and $\ell = \ell' + 1$. It can be easily verified that $A(T_{i+1}) = A(T_i) \cup \{u, v, w\}$ is a 3-component dominating set. Thus $\gamma_3(T) \leq \gamma_3(T') + 3$. From observation 1 and observation 3, $\{v, w\} \subseteq D$. Combining with observation 2, let us assume $u \in D$. It is clear that u has no neighbors in D except for w , otherwise, we can easily obtain a smaller 3-component dominating set, which contradicts the assumption that D is a $\gamma_3(T)$ -set. Therefore, $D \setminus \{u, v, w\}$ is a 3-component dominating set of T' , that is, $\gamma_3(T') \leq \gamma_3(T) - 3$. Consequently, we can conclude that $\gamma_3(T) = \gamma_3(T') + 3 = \frac{3(n-\ell+2)}{5}$. $\square$

3. Results on the generalized Petersen graphs

In this section, we will give the exact values of the 3-component domination number of $P(n, k)$, where $k = 1, 2$. Before this, we first give the following useful lemma.

Lemma 7. Let $n \geq 2k + 1$. Then $\gamma_s(P(n, k)) \geq \lceil \frac{sn}{s+1} \rceil$.

Proof. Let $G = P(n, k)$ with $n \geq 2k + 1$. Suppose D is a minimal s -component dominating set of G , and let K_1 be a component of $G[D]$ of order t . Since G is a cubic graph, the total degree of $V(K_1)$ is $\sum_{v \in K_1} d_G(v) = 3t$. Moreover, the edges between K_1 and $G \setminus K_1$ are at most $3t - 2(t - 1) = t + 2$, because $G[K_1]$ is connected and has at least $t - 1$ edges. Therefore, the maximum number of vertices that K_1 can dominate, including $V(K_1)$ itself, is at most $t + (t + 2) = 2t + 2$.

Now assume that there are m components in $G[D]$ with each component of order t_i for $1 \leq i \leq m$. The maximum number of vertices that D can dominate is then given by $\sum_{i=1}^m (2t_i + 2) = 2|D| + 2m$. From the definition of s -component domination, $t_i \geq s$ for all i . Thus $m \leq \lfloor \frac{|D|}{s} \rfloor$. Since D is also a dominating set, $2|D| + 2\lfloor \frac{|D|}{s} \rfloor \geq 2|D| + 2m \geq 2n$, which implies the conclusion. $\square$

Corollary 8. $\gamma_s(P(n, 1)) = \frac{sn}{s+1} + o(1)$.

Proof. From Lemma 7, we know that $\gamma_s(P(n, k)) \geq \lceil \frac{sn}{s+1} \rceil$, or equivalently, $\gamma_s(P(n, 1)) \geq \frac{sn}{s+1} + o(1)$. To establish the upper bound, we construct a 3-component dominating set S as follows. If $n \equiv j \pmod{(2s+2)}$, where $0 \leq j \leq 2s+1$, then we select the set S as $\{u_{(2s+2)i+1}, u_{(2s+2)i+2}, \dots, u_{(2s+2)i+s}\} \cup \{v_{(2s+2)i+(s+2)}, v_{(2s+2)i+(s+3)}, \dots, v_{(2s+2)i+(2s+1)}\}$ over $0 \leq i \leq \lfloor \frac{n}{2s+2} \rfloor - 1$ and add $\{u_n, u_{n-1}, \dots, u_{n-j+1}\}$ to the set. Obviously S is a 3-component dominating set of $P(n, 1)$, therefore, $\gamma_s(P(n, k)) \leq \frac{sn}{s+1} + o(1)$. This completes the proof. $\square$

Now we state our first result.

Theorem 9.

$$\gamma_3(P(n, 1)) = \begin{cases} \lceil \frac{3n}{4} \rceil, & n \equiv 0, 1, 2, 3, 7 \pmod{8}; \\ \lceil \frac{3n}{4} \rceil + 1, & \text{otherwise.} \end{cases}$$

Proof. For the upper bound, we construct a set S as follows. Let $n \equiv 8\ell + j$, where $0 \leq j \leq 7$ and $i \in [0, \ell - 1]$ are two integers. First, we select vertices $\{u_{8i+1}, u_{8i+2}, u_{8i+3}\} \cup \{v_{8i+5}, v_{8i+6}, v_{8i+7}\}$ for all i . Then we add additional vertices to S depending on the value of j . If $1 \leq j \leq 6$, then add to S $\{u_n, u_{n-1}, \dots, u_{n-j+1}\}$. If $j = 7$, then add to S $\{u_{n-6}, u_{n-5}, u_{n-4}, v_{n-3}, v_{n-2}, v_{n-1}\}$. It can be verified that S is a 3-component dominating set, thereby yielding the upper bound as stated in the theorem. For instance, Fig. 2 depicts the case of $P(8, 1)$ as an example.

The result is valid for $n \equiv 0, 1, 2, 3, 7 \pmod{8}$, as $\gamma_3(P(n, 1)) \geq \lceil \frac{3n}{4} \rceil$ from Lemma 7. Now we will prove the theorem for the remaining cases, assuming $G = P(n, 1)$ for convenience. We will consider the case when $n = 8\ell + 4$ and the remaining cases follow the same approach.

Suppose that the theorem does not hold, G contains a 3-component dominating set D' of order $\lceil \frac{3n}{4} \rceil = 6\ell + 3$. According to the details presented in [Lemma 7](#), there can be at most $\lfloor \frac{|D'|}{3} \rfloor = 2\ell + 1$ components in D' . Therefore, D' can dominate at most $2(6\ell + 3 + 2\ell + 1) = 16\ell + 8 = 2n$ vertices. Since D' is also a dominating set, it must dominate exactly $2n$ vertices. Hence, the following facts hold.

- (1) Each component of $G[D']$ consists of exactly 3 vertices.
- (2) The vertices not in D' are dominated precisely once by D' .

Let K be any component of $G[D']$. Recall that $P(n, 1)$ has vertex sets $U = \{u_1, u_2, \dots, u_n\}$ and $V = \{v_1, v_2, \dots, v_n\}$. We can now state the following claims.

Claim 1. $V(K) \subset U$ or $V(K) \subset V$.

Proof of Claim 1. Assume the contrary, that there exist two vertices $u_i \in U$ and $v_i \in V$ that belong to the same component K . Since $|K| = 3$, only one of $\{u_{i+1}, u_{i-1}, v_{i+1}, v_{i-1}\}$ belongs to $V(K)$. Without loss of generality, suppose $u_{i-1} \in V(K)$. Then v_{i-1} is dominated by both u_{i-1} and v_i , which contradicts the fact (2). $\square$

Claim 2. For any $i \in [2\ell + 1]$, K_i and K_{i+1} are different types. This means if K_i is an U -component, then K_{i+1} is a V -component.

Proof of Claim 2. Assume the contrary, that $K_1 = \{u_1, u_2, u_3\}$ and $K_2 = \{u_i, u_{i+1}, u_{i+2}\}$ are two U -components. Then $i \geq 6$, or otherwise, u_4 is dominated by K_1 and K_2 , which contradicts the fact (2). However, the vertices $\{v_4, v_5\}$ cannot be dominated by any component in $G[D']$ for $i \geq 6$, a contradiction. $\square$

Let us construct an auxiliary cycle graph C as follows: each component K_i ($i \in [2\ell + 1]$) of $G[D']$ is considered as a vertex v_i of C , and the edge set consist of $\{v_i v_{i+1}\}$, where subscripts are taken modulo $2\ell + 1$. This creates a cycle C of length $2\ell + 1$. According to [Claim 2](#), consecutive vertices in C correspond to different types of components in G . However, this is impossible because C is an odd cycle. Therefore, we conclude that $\gamma_3(G) > \lceil \frac{3n}{4} \rceil$, which implies $\gamma_3(G) \geq \lceil \frac{3n}{4} \rceil + 1$. Thus, the theorem holds. $\square$

Next, we give the exact value for $\gamma_3(P(n, 2))$.

Theorem 10.

$$\gamma_3(P(n, 2)) = \begin{cases} \lceil \frac{3n}{4} \rceil, & n \equiv 0, 2, 3, 5, 7 \pmod{8}; \\ \lceil \frac{3n}{4} \rceil + 1, & \text{otherwise.} \end{cases}$$

To prove [Theorem 10](#), we will first establish the upper bounds and then provide three lemmas to establish the lower bounds.

Proof. For the upper bounds. We construct a 3-component dominating set S as follows. Let $n \equiv 8\ell + j$, where $0 \leq j \leq 7$ and $i \in [0, \ell - 1]$ are two integers.

First, we select $\{u_{8i+1}, u_{8i+2}, v_{8i+2}\} \cup \{u_{8i+5}, u_{8i+6}, v_{8i+5}\}$ for all i , then add extra vertices to S based on the value of j .

If $j = 1$, then add $\{u_{n-9}, u_{n-8}, u_{n-7}, v_{n-8}\} \cup \{u_{n-4}, u_{n-3}, u_{n-2}, v_{n-3}\}$ to S .

If $j = 2$, then add $\{u_{n-8}, u_{n-7}, u_{n-6}, v_{n-7}\} \cup \{u_{n-4}, u_{n-3}, u_{n-2}, v_{n-3}\}$ to S .

If $j = 3$, then add $\{u_{n-3}, u_{n-2}, u_{n-1}\}$ to S .

If $j = 4$, then add $\{u_{n-4}, u_{n-3}, u_{n-2}, u_{n-1}\}$ to S .

If $j = 5$, then add $\{u_{n-4}, u_{n-3}, u_{n-2}, v_{n-3}\}$ to S .

If $j = 6$, then add $\{u_{n-5}, u_{n-4}, u_{n-3}, u_{n-2}, v_{n-3}, v_{n-4}\}$ to S .

If $j = 7$, then add $\{u_{8\ell+1}, u_{8\ell+2}, v_{8\ell+2}\} \cup \{u_{8\ell+5}, u_{8\ell+6}, v_{8\ell+5}\}$ to S .

It is easy to check that S is a 3-component dominating set, thus

$$\gamma_3(P(n, 2)) \leq \begin{cases} \lceil \frac{3n}{4} \rceil, & n \equiv 0, 2, 3, 5, 7 \pmod{8}; \\ \lceil \frac{3n}{4} \rceil + 1, & \text{otherwise.} \end{cases}$$

As stated in [Lemma 7](#), we learn that $\gamma_3(P(n, 2)) \geq \lceil \frac{3n}{4} \rceil$. Hence, the theorem holds when $n \equiv 0, 2, 3, 5, 7 \pmod{8}$. We will establish the lower bounds for $n \equiv 1, 4, 6 \pmod{8}$ by the following three lemmas to complete this proof. $\square$

Lemma 11. $\gamma_3(P(n, 2)) \geq \lceil \frac{3n}{4} \rceil + 1$ for $n \equiv 1 \pmod{8}$.

Proof. We prove this lemma by contradiction. Let $n = 8\ell + 1$. Suppose there is a 3-component dominating set D' of order $\lceil \frac{3n}{4} \rceil = 6\ell + 1$. As proved in [Lemma 7](#), we obtain the following observations: (i) There are at most $\lfloor \frac{|D'|}{3} \rfloor = 2\ell$ components, thus D' can dominate at most $2(6\ell + 1 + 2\ell) = 16\ell + 2 = 2n$ vertices. (ii) Since D' is also a dominating set, D' dominate exactly $2n$ vertices. In what follows, three facts are given:

- Fact (1): There are no components in D' with the order greater than 4.
 Fact (2) All vertices not in D' are dominated exactly once.
 Fact (3): All components in $G[D']$ are trees.

Proof of the facts.

- (1) Following the same idea presented in [Lemma 7](#), we can conclude that no component has an order greater than 5. Assume there are x components of order 3, y components of order 4, and z components of order 5. Then solve the following functions:

$$x + y + z = 2\ell \quad (1)$$

$$3x + 4y + 5z = 6\ell + 1 \quad (2)$$

$$x, y, z \in \mathbb{N} \quad (3)$$

Hence, we obtain that $x = 2\ell - 1, y = 1, z = 0$.

- (2) Assume there exists a vertex v outside $V(D')$, which is dominated by two components of $G[D']$, namely K_1 and K_2 . According to the previous analysis, each component $K \subset G[D']$ can dominate at most $2|K| + 2$ vertices. Based on this analysis, it follows that any two components K_i and K_j together dominate exactly $2(|K_1| + |K_2|) + 4$ vertices. However, the union $K_1 \cup K_2$ can dominate at most $2(|K_1| + |K_2|) + 3$ vertices.
- (3) If a cycle exists in the component $K_1 \subset G[D']$, it can be easily verified that K_1 can dominate at most $2|K_1|$ vertices. However, according to the previous assumption, every component K must dominate exactly $2|K| + 2$ vertices, which leads to a contradiction. $\square$

Observation 12. The distance between any two closest components is exactly 3.

Proof of Observation 12. Suppose there exist two closest components K_1 and K_2 whose distance is not 3, and let P be the corresponding shortest path. If $d(K_1, K_2) \geq 4$, set $P = w_0 w_1 w_2, \dots, w_x$ with $w_0 \in V(K_1)$ and $w_x \in V(K_2)$, where $x \geq 4$. Since D' is a dominating set, then there exists $i \neq 1, 2$ and a vertex $v' \in V(K_i)$ that v' dominates w_2 . This means that $d(K_1, K_i) < d(K_1, K_2)$, which leads to a contradiction. If $d(K_1, K_2) = 2$, set $P = w_0 w_1 w_2$, where $w_0 \in V(K_1)$ and $w_2 \in V(K_2)$. In this case, w_1 is dominated twice, which contradicts the Fact (3). $\square$

To complete the proof, we will show that D' is not a 3-component dominating set of G . Based on the previous analysis, we can conclude that there is only one component of order 4, denoted as K' . Then all possible constructions of K' are described below:

- (1) $G[K'] = G[\{u_{i-4}, u_{i-3}, u_{i-2}, u_{i-1}\}]$.
- (2) $G[K'] = G[\{v_{i-3}, v_{i-1}, v_{i+1}, v_{i+3}\}]$.
- (3) $G[K'] = G[\{u_{i-4}, u_{i-3}, u_{i-2}, v_{i-4}\}]$.
- (4) $G[K'] = G[\{v_{i-3}, v_{i-1}, v_{i+1}, u_{i-3}\}]$.
- (5) $G[K'] = G[\{v_{i-3}, u_{i-3}, u_{i-2}, v_{i-2}\}]$.
- (6) $G[K'] = G[\{u_{i-3}, u_{i-2}, u_{i-1}, v_{i-2}\}]$.
- (7) $G[K'] = G[\{v_{i-3}, v_{i-1}, v_{i+1}, u_{i-1}\}]$.
- (8) $G[K'] = G[\{v_{i-3}, v_{i-1}, u_{i-1}, u_i\}]$.

Next, we prove that none of these constructions exist.

- (1) [Observation 12](#) implies that none of the vertices $\{v_i, u_{i+1}, v_{i+1}\}$ belongs to D' . In order to dominate these vertices, the vertices $\{v_{i+2}, u_{i+2}, v_{i+3}\}$ should be included in D' . However, since there is only one 4-component in $G[D']$, the vertices $\{v_{i+2}, u_{i+2}, v_{i+3}\}$ belong to two different components whose distance is of 2, which contradicts [observation 12](#).
- (2) Following the proof in (1), we find that $\{v_{i-2}, v_i, v_{i+2}\} \subset D'$. In order to dominate $u_{i-4}, u_{i-5} \in D'$, since none of $\{v_{i-4}, u_{i-3}\}$ belongs to D' . However, v_{i-5} is dominated twice, which contradicts the fact (3).
- (3) Similarly, it is easy to check that v_{i-2} is dominated by v_{i-4} and u_{i-2} , a contradiction.
- (4) [Observation 12](#) implies that the vertices in $S = \{u_i, u_{i+2}, u_{i+3}\}$ cannot belong to D' . In order to dominate the vertices in S , we need to add $\{v_i, v_{i+2}, v_{i+4}, u_{i+4}\}$ to D' , which forms a 4-component of D' . However, this is impossible because that no two 4-components in $G[D']$.
- (5) Similarly, we can deduce that $\{u_i, v_{i+1}, v_{i+2}\}$ cannot belong to D' . This implies that $\{u_{i+1}, u_{i+2}, u_{i+3}, v_{i+3}\}$ forms another 4-component in $G[D']$, which is a contradiction.

Now let us consider the remaining cases, which follow a similar proof idea. We may assume that the next component closest to K' is denoted as K_1 , and K_2 is the subsequent component closest to K_1 , and $K_3, \dots, K_{2\ell-1}$ defined in the same way. In the following proof, some kinds of components are utilized and illustrated in [Fig. 3](#), labeled from types A to H.

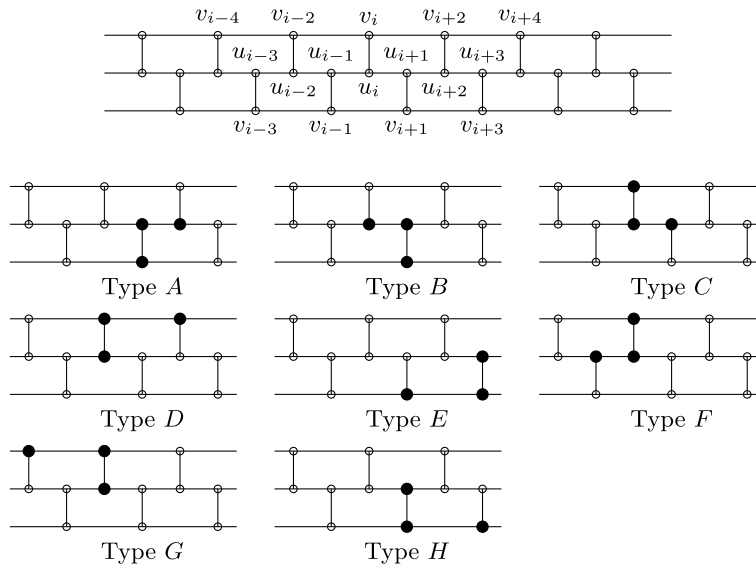


Fig. 3. Type A to Type H and Black vertices are chosen to be dominating vertices.

- (6) According to the previous analysis, we can conclude that $K_1 = \{u_{i+2}, u_{i+3}, v_{i+3}\}$ is type B, and $K_{2\ell-1} = \{u_{i-7}, u_{i-6}, v_{i-7}\}$ is type A. Now let us study the construction of K_2 . By the definition of 3-component dominating set and the structure of $P(n, 2)$, either $K_2 = \{u_{i+6}, v_{i+6}, u_{i+7}\}$ is type C or $K_2 = \{u_{i+6}, v_{i+6}, v_{i+8}\}$ is type D. If K_2 is type C, then we can deduce that $K_3 = \{u_{i+10}, u_{i+11}, v_{i+11}\}$ is type B. On the other hand, if K_2 is type D, then $K_3 = \{v_{i+9}, v_{i+11}, u_{i+11}\}$ is type E. Employing the same analysis, we obtain there are several constructions shown in Fig. 3, and every component in $G[D']$ belongs to one of them. Let “ $X \rightarrow Y$ ” means that if K_m is type X, then K_{m+1} is type Y and “ $X \leftarrow Y$ ” is defined similarly. So we can obtain $C \rightarrow B, D \rightarrow E, E \rightarrow C$ or $E \rightarrow D, B \rightarrow C$ or $B \rightarrow D$. Therefore we conclude $K_i \in \{B, C, D, E\}$ over $1 \leq i \leq 2\ell - 1$. However, $K_{2\ell-1}$ is type A, which yields a contradiction.
- (7) The proof follows the same idea as in case (6). In this case, K_1 is type G, $K_{2\ell-1}$ is type D, and the subsequent rule is followed for the same reason.

$$A \rightarrow F \rightarrow \begin{cases} A \\ H \rightarrow G \rightarrow \begin{cases} A \\ H \end{cases} \end{cases}$$

Again we obtain $K_i \in \{A, F, G, H\}$ over $1 \leq i \leq 2\ell - 2$, none of them can lead to $K_{2\ell-1}$, a contradiction.

- (8) Now K_1 is type F, then $K_i \in \{A, F, G, H\}$ over $1 \leq i \leq 2\ell - 2$ for the same reason. However, $K_{2\ell-1}$ is type D, again we obtain that none of them can lead to $K_{2\ell-1}$, a contradiction. $\square$

Lemma 13. $\gamma_3(P(n, 2)) \geq \lceil \frac{3n}{4} \rceil + 1$ for $n \equiv 4 \pmod{8}$.

Proof. Assume the contrary, let $\gamma_3(P(n, 2)) = \lceil \frac{3n}{4} \rceil$ for $n = 8\ell + 4$ and suppose there is a 3-component dominating set D' of order $\lceil \frac{3n}{4} \rceil = 6\ell + 3$. From the proof of Lemma 7, we know that there are at most $\lfloor \frac{|D'|}{3} \rfloor = 2\ell + 1$ components. Hence, D' can dominate at most $2(6\ell + 3 + 2\ell + 1) = 16\ell + 8 = 2n$ vertices. Similarly, the following facts hold:

Fact (4): Every component of D' is of order 3, and there are $2\ell + 1$ components in total.

Fact (5): All the components in $G[D']$ are trees.

Fact (6): All the vertices not in D' are dominated exactly once.

Fact (7): Observation 12 still holds.

We subsequently prove the following observation.

Observation 14. No component in $G[D']$ consists of vertices all from U or V .

Proof of Observation 14. Assuming the contrary, let us first consider the case where $\{u_{i-1}, u_i, u_{i+1}\} \subseteq G[D']$. Observation 12 implies that none of $\{v_{i+2}, u_{i+2}, v_{i+3}\}$ belongs to D' . However, to dominate these vertices, the vertices $\{v_{i+4}, u_{i+4}, v_{i+5}\}$ must belong to D' . Since all the components in $G[D']$ are of order 3 by fact (4), they cannot be in the

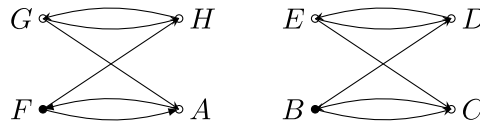


Fig. 4. The auxiliary graph.

same component. Therefore, u_{i+4}, v_{i+4} belong to the same component K , while v_{i+5} belongs to another component K' . However, the distance between K and K' is at most 2, which contradicts the assumption.

Next, if $\{v_{i-2}, v_i, v_{i+2}\} \subseteq G[D']$, then none of $\{u_{i-1}, u_{i+1}, u_{i+3}, u_{i+4}\}$ belongs to D' by observation 12. To dominate these vertices, $\{v_{i-1}, v_{i+1}, v_{i+3}, u_{i+5}\} \subseteq G[D']$. However, this leads to a contradiction because the component containing u_{i+5} is at a distance of at most 2 from $\{v_{i-1}, v_{i+1}, v_{i+3}\}$. $\square$

By observation 14, Fig. 3 displays all possible constructions of components of order 3. Based on the definition and the structure of G , we can deduce the following rules:

$$A \rightarrow F \rightarrow \begin{cases} A \\ H \end{cases} \rightarrow G \rightarrow \begin{cases} A \\ H \end{cases}$$

$$C \rightarrow B \rightarrow \begin{cases} C \\ D \end{cases} \rightarrow E \rightarrow \begin{cases} C \\ D \end{cases}$$

Therefore, the following two conclusions hold for exactly one:

$K_i \in \{B, C, D, E\}$ over $1 \leq i \leq 2\ell + 1$, or

$K_i \in \{A, F, G, H\}$ over $1 \leq i \leq 2\ell + 1$.

We construct a digraph with vertices set $\{B, C, D, E\}$, and edge set $E = \{C \rightarrow B, D \rightarrow E, E \rightarrow C, E \rightarrow D, B \rightarrow C, B \rightarrow D\}$ (refer to Fig. 4). The problem now reduces to finding a cycle of length 3. However, this is impossible since the graph is bipartite. Therefore, the desired result holds.

Lemma 15. $\gamma_3(P(n, 2)) \geq \lceil \frac{3n}{4} \rceil + 1$ for $n \equiv 6 \pmod{8}$.

Proof. Suppose the contrary, assuming there is a 3-component dominating set D' of order $\lceil \frac{3n}{4} \rceil = 6\ell + 5$. From the proof of Lemma 7, we can deduce the following facts:

Fact (8): Every component of D' is of order at most 5, and there are $2\ell + 1$ components in total.

Fact (9): All the components in $G[D']$ are trees.

Fact (10): Each vertex not in D' is dominated only once.

Fact (11): Observation 12 still holds.

Based on the fact (8), we can classify the cases according to the maximal order of the components:

(i) One component has a order of 5, and the remaining 2ℓ components have a order of 3.

(ii) Two components have a order of 4, and the remaining $2\ell - 1$ components have a order of 3.

To complete the proof, we will show that none of these cases exist.

Proof of case. (i): We will only consider the two possible constructions, as the proofs of other possible constructions are the same as in Lemma 13. Denote such component as K_1 .

(1) $K_1 = \{v_i, v_{i+2}, v_{i+4}, v_{i+6}, u_i\}$.

According to observation 12, we know that $\{v_{i+3}, v_{i+5}, v_{i+7}, u_{i+9}\} \subseteq D'$. However, this is impossible because the component formed by $\{v_{i+3}, v_{i+5}, v_{i+7}\}$ is at a distance of 2 from the component containing u_{i+9} .

(2) $K_1 = \{v_i, v_{i+2}, v_{i+4}, v_{i+6}, u_{i+4}\}$.

Again, it can be obtained that $\{v_{i-3}, v_{i-1}, v_{i+1}, u_{i-3}\}$ forms a 4-component, leading to a contradiction. $\square$

Before the proof of case (ii), we need the following claim.

Claim. Observation 14 still holds.

Proof of the claim. Let us assume that $K_i = \{u_{i-1}, u_i, u_{i+1}\} \subseteq D'$ (type I in Fig. 5). Then we deduce that $K_{i+1} = \{v_{i+4}, u_{i+4}, u_{i+5}, v_{i+5}\} \subseteq D'$ (type J) and $K_{i-1} = \{v_{i-4}, u_{i-4}, u_{i-5}, v_{i-5}\} \subseteq D'$ (type L). Therefore, we obtain $I \rightarrow J$ and $I \leftarrow L$. It can be easily verified that $J \rightarrow I$, which means that if K_{i+1} is type J, then $K_j \in \{I, J\}$ over $j \in [2\ell - 1]$. However, K_{i-1} is type L, which leads to a contradiction.

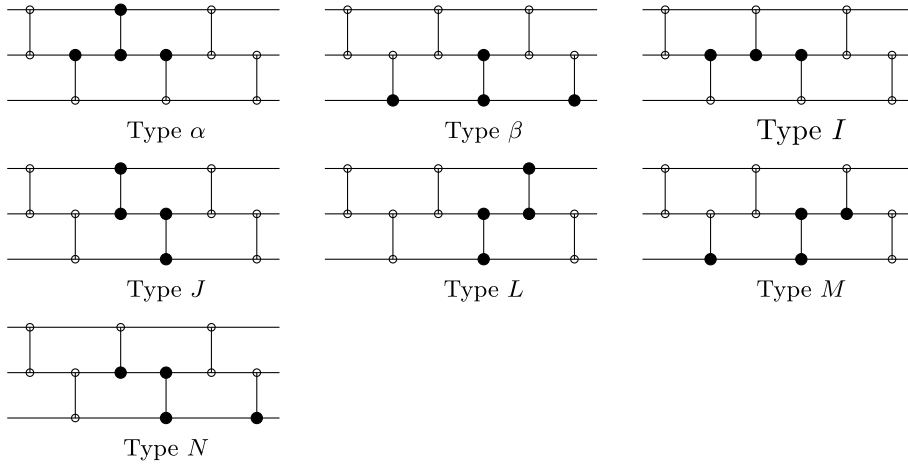


Fig. 5. Type α to Type N and Black vertices are chosen to be dominating vertices.

Now assuming $\{v_{i-3}, v_{i-1}, v_{i+1}, v_{i+3}\} \subseteq D'$, then definitely, we have $\{v_{i-4}, v_{i-2}, v_i, v_{i+2}, v_{i+4}\} \subseteq D'$, which forms a 5-component, which contradicts the assumption. $\square$

Proof of case (ii): Let K_x and K_y be the two 4-components. As before, we consider all possible constructions of 4-component, without loss of generality, we let K' be a 4-component, and $K_{2\ell} \leftarrow K' \rightarrow K_1$:

- (1) $K' = \{u_{i-4}, u_{i-3}, u_{i-2}, u_{i-1}\}$.

We can obtain that none of $\{v_i, u_{i+1}, v_{i+1}\}$ belongs to D' . Therefore, $\{v_{i+2}, u_{i+2}, v_{i+3}\} \subset D'$. If these vertices belong to two different components, it would contradict observation 12 since the distance between these two components is 2. Hence, $\{v_{i+2}, u_{i+2}, v_{i+3}\}$ forms a 4-component. Similarly, we can deduce that $\{v_{i-8}, u_{i-8}, u_{i-7}, v_{i-7}\}$ is also a 4-component. However, there are only two 4-components, leading to a contradiction.

- (2) $K' = \{v_{i-3}, v_{i-1}, v_{i+1}, v_{i+3}\}$.

We can obtain that none of $\{u_{i-5}, u_{i-4}, u_{i-2}, u_i, u_{i+2}, u_{i+4}, u_{i+5}\}$ belongs to D' . To dominate them, $\{v_{i-4}, v_{i-2}, v_i, v_{i+2}, v_{i+4}\} \subseteq D'$, which forms a 5-component, which is a contradiction.

- (3) $K' = \{u_{i-4}, u_{i-3}, u_{i-2}, v_{i-4}\}$.

The vertex v_{i-2} is dominated by v_{i-4} and u_{i-2} , which is a contradiction.

- (4) $K' = \{v_{i-3}, v_{i-1}, v_{i+1}, u_{i-3}\}$.

We can obtain $K_1 = \{v_i, v_{i+2}, v_{i+4}, u_{i+4}\}$, which leads to K_2 is either type A or type H . Moreover, $K_{2\ell}$ is either type G or F . As shown before, we can construct a digraph with vertex set $\{A, F, G, H\}$, and edge set $\{A \rightarrow F, H \rightarrow G, G \rightarrow A, G \rightarrow H, F \rightarrow A, F \rightarrow H\}$. Therefore, the problem reduces to finding an odd-length path from A (or H) to G (or F). However, it can be easily proven that no such paths exist in this digraph, which is a contradiction.

- (5) $K' = \{v_{i-3}, u_{i-3}, u_{i-2}, v_{i-2}\}$.

The proof is the same as in Lemma 15.

- (6) $K' = \{u_{i-4}, u_{i-3}, u_{i-2}, v_{i-3}\}$.

- (7) $K' = \{v_{i-3}, v_{i-1}, v_{i+1}, u_{i-1}\}$.

- (8) $K' = \{v_{i-3}, v_{i-1}, u_{i-1}, u_i\}$.

The remaining cases of (6)-(8) are corresponding the type α , β and M or N (the dual case of (7)). We deduce the following rules based on a similar analysis as before: $A \leftarrow \alpha \rightarrow B$, $D \leftarrow \beta \rightarrow G$ and $D \leftarrow M \rightarrow F$, and $C \leftarrow N \rightarrow G$.

Now we aim to find the construction: $X \rightarrow x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_{m_1} \rightarrow Y \rightarrow y_1 \rightarrow y_2 \rightarrow \dots \rightarrow y_{m_2} \rightarrow X$, where $X, Y \in \{\alpha, \beta, M, N\}$ and x_i, y_i belongs to exactly one of $\{B, C, D, E\}$ or $\{A, F, G, H\}$.

Since there are $2\ell - 1$ components, thus either m_1 or m_2 is an odd number. Again we construct two digraphs: G_1 with vertex set $\{B, C, D, E\}$, edge set $E_1 = \{C \rightarrow B, D \rightarrow E, E \rightarrow C, E \rightarrow D, B \rightarrow C, B \rightarrow D\}$, and G_2 with vertex set $\{A, F, G, H\}$, edge set $E_2 = \{A \rightarrow F, H \rightarrow G, G \rightarrow A, G \rightarrow H, F \rightarrow A, F \rightarrow H\}$. We need to find an odd-length path in G_1 or G_2 . Observing G_1 , we find that there are only two such paths: from B to E or from E to B . Similarly, an odd path in G_2 is either the path from F to G or the path from G to F . Without loss of generality, let X be type α , then by the rule $A \leftarrow \alpha \rightarrow B$, if m_1 is odd, then $x_{m_1} = E \rightarrow Y$, which implies that Y must be type C or type D . However, $Y \in \{\alpha, \beta, M, N\}$, which is a contradiction. The remaining cases for $X \in \{\beta, M, N\}$ can be proven similarly.

Combining the above, the lemma follows. $\square$

Now we conclude that Theorem 10 holds. Furthermore, using the same idea, we provide the following result.

Theorem 16.

$$\gamma_4(P(n, 2)) = \begin{cases} \lceil \frac{4n}{5} \rceil, & n \equiv 0 \text{ or } 4 \pmod{5}; \\ \lceil \frac{4n}{5} \rceil + 1, & \text{otherwise.} \end{cases}$$

Proof. We give the following constructions to establish the upper bound. Let $n \equiv 5\ell + j$, where $0 \leq j \leq 4$ and $i \in [0, \ell - 1]$ are two integers.

First, select $\{u_{5i+2}, u_{5i+3}, u_{5i+4}\} \cup \{v_{5i+3}\}$ for all i . And add additional vertices to S depending on the value of j .

If $j = 1$, then add to S $\{u_{n-5}, u_{n-4}, u_{n-3}, u_{n-2}, u_{n-1}, u_n\}$.

If $j = 2$, then add to S $\{u_{n-6}, u_{n-5}, u_{n-4}, u_{n-3}, u_{n-2}, u_{n-1}, u_n\}$.

If $j = 3$, then add to S $\{u_{n-7}, u_{n-6}, u_{n-5}, u_{n-4}, u_{n-3}, u_{n-2}, u_{n-1}, u_n\}$.

If $j = 4$, then add to S $\{u_{5\ell+2}, u_{5\ell+3}, u_{5\ell+4}\} \cup \{v_{5\ell+3}\}$.

The proof of lower bounds is similar to [Theorem 10](#), we omit it. $\square$

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Data availability

No data was used for the research described in the article.

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